tonemachine

Project Report

1. Readme

ToneMachine

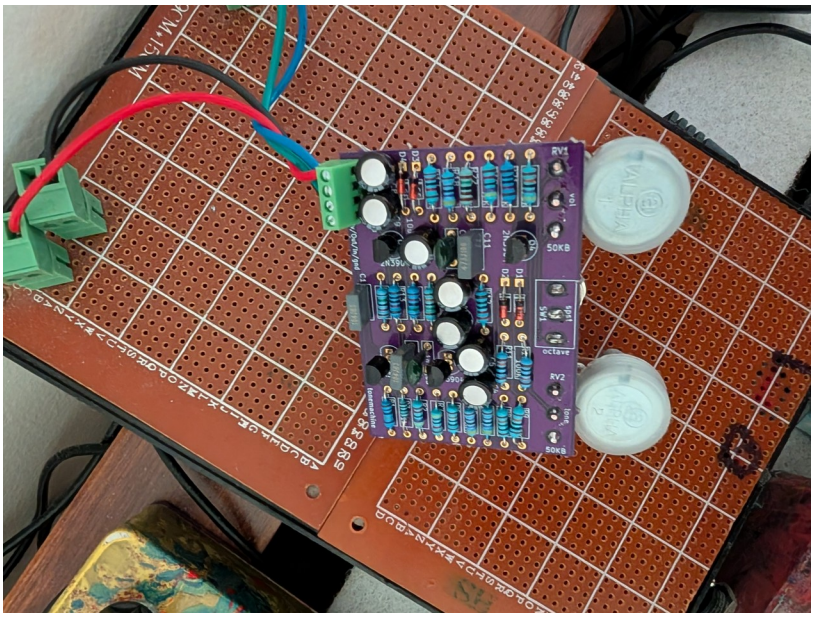
Clone based on Foxx Tone machine

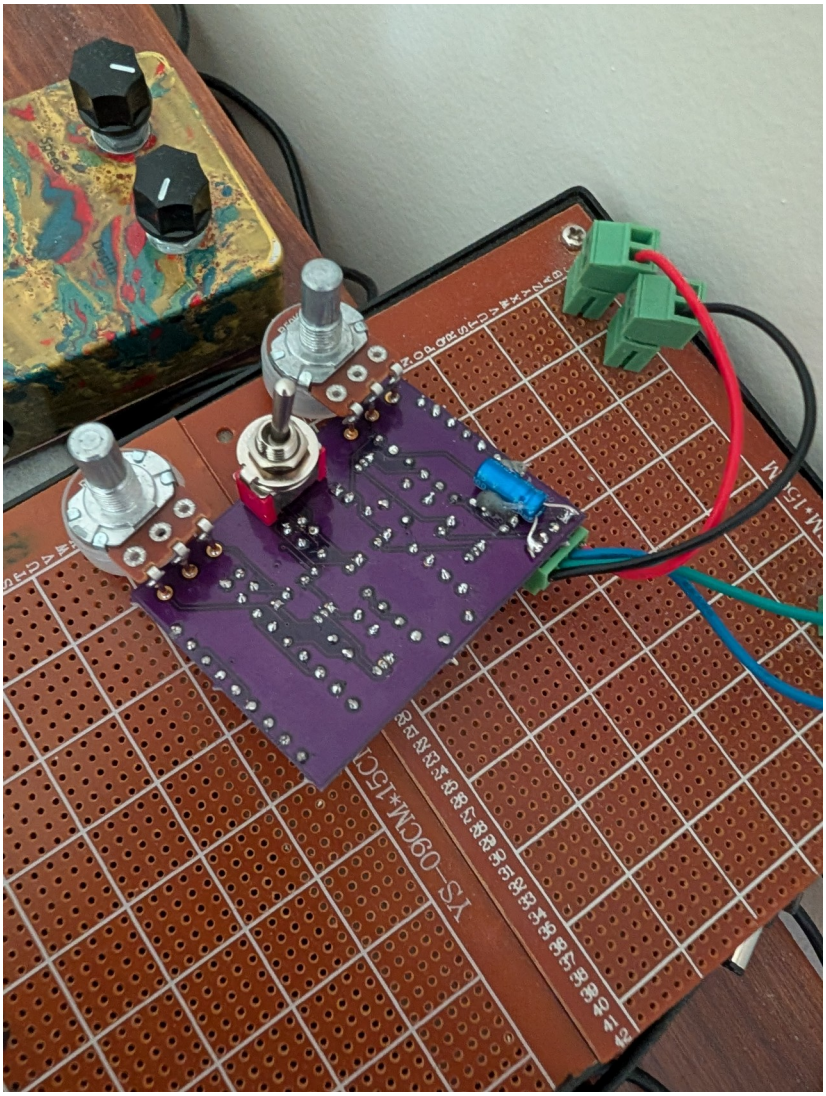
Changed font size, going to try to run these at jlpcb, and the size I've been using is below their minimum.

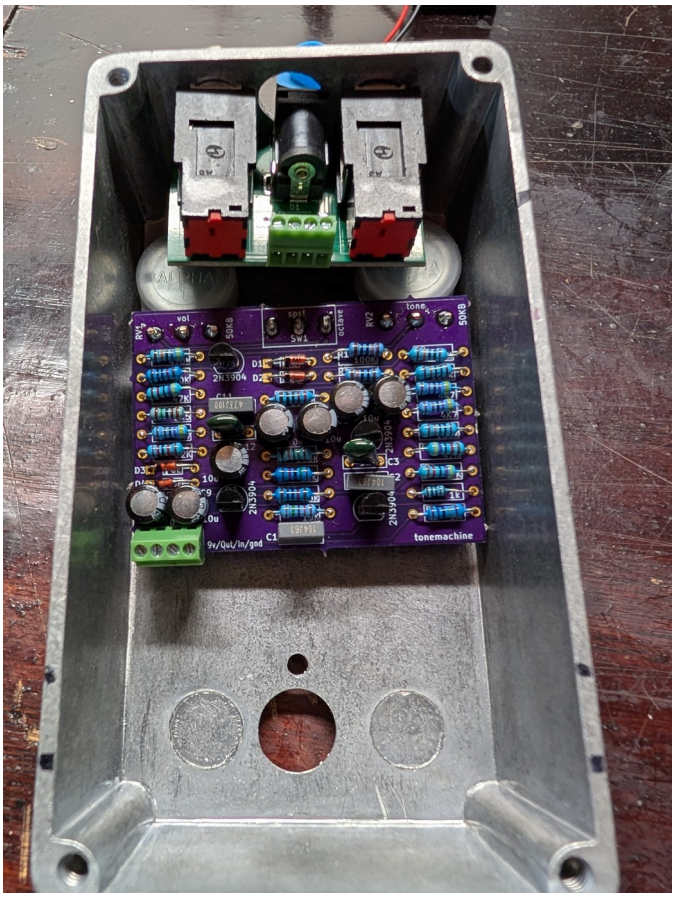
Original schematic used did not include a 100u electrolytic on the 9v line, but the tagboardeffects build did, so I added this into the layout. I bodged this onto the underside of the board.

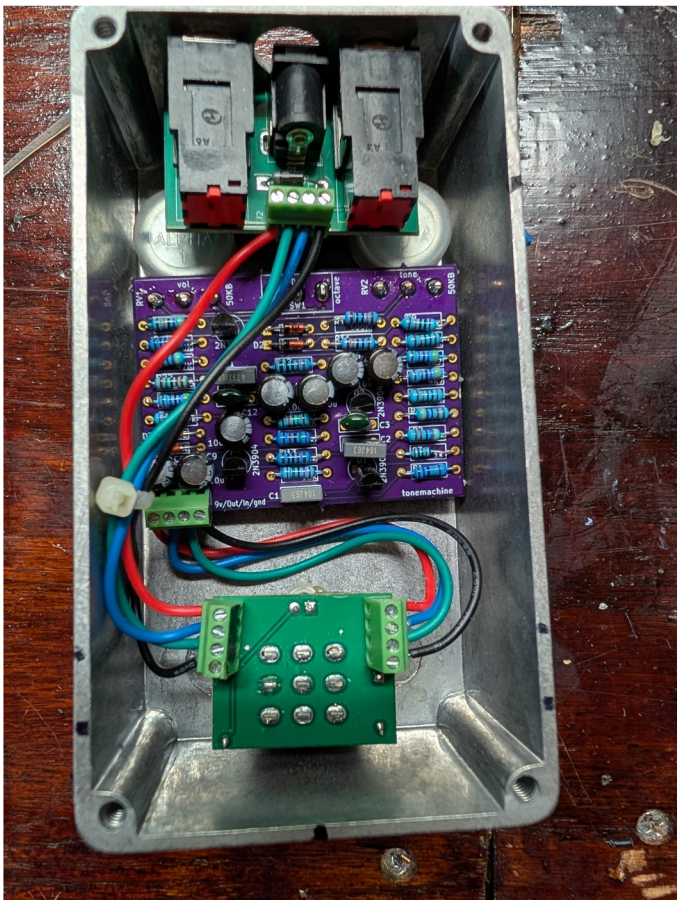
The transistors used in the build were 2n3904. And for the germanium diodes, BAT42 were used.

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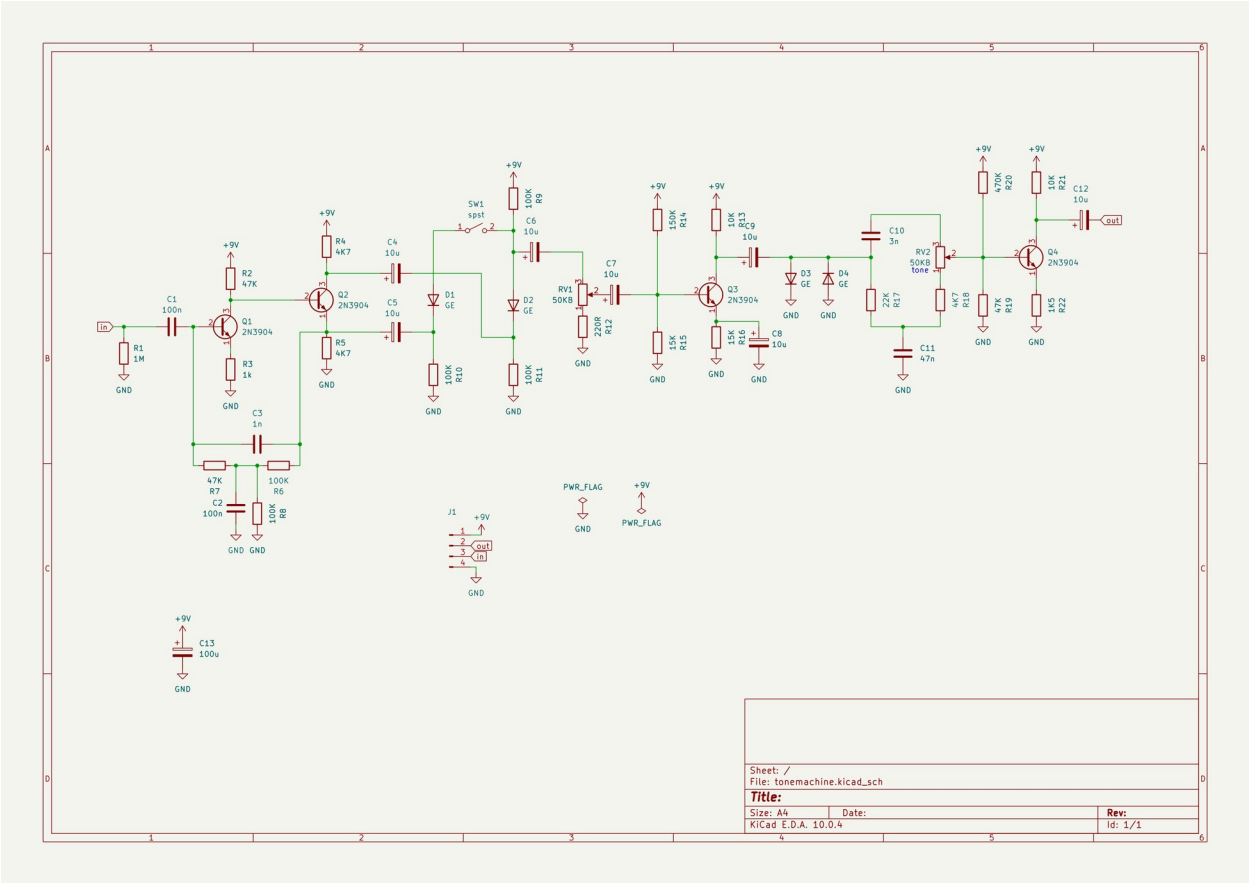

RV1
50KB
1MEG
octave
RV2
50KB



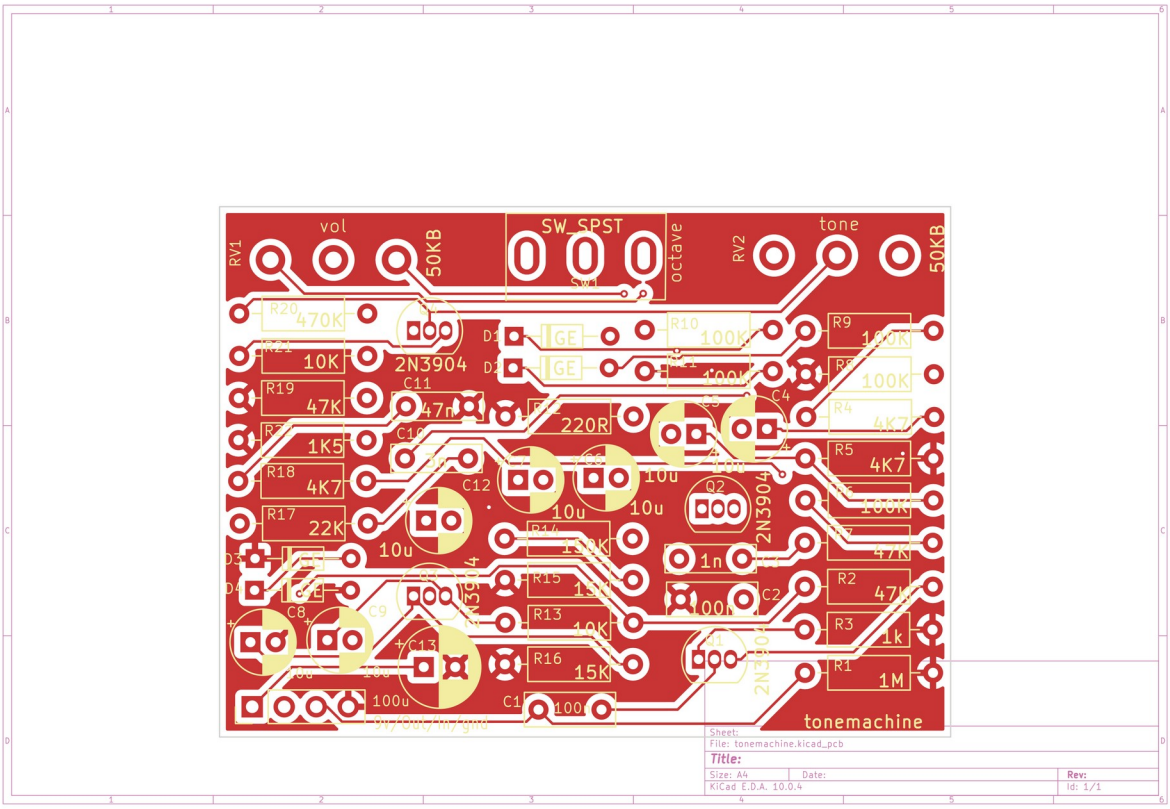

vol
tone
octave
tonemachine
9v/Out/in/gnd


vol
tone
octave
tonemachine
9v/Out/in/gnd

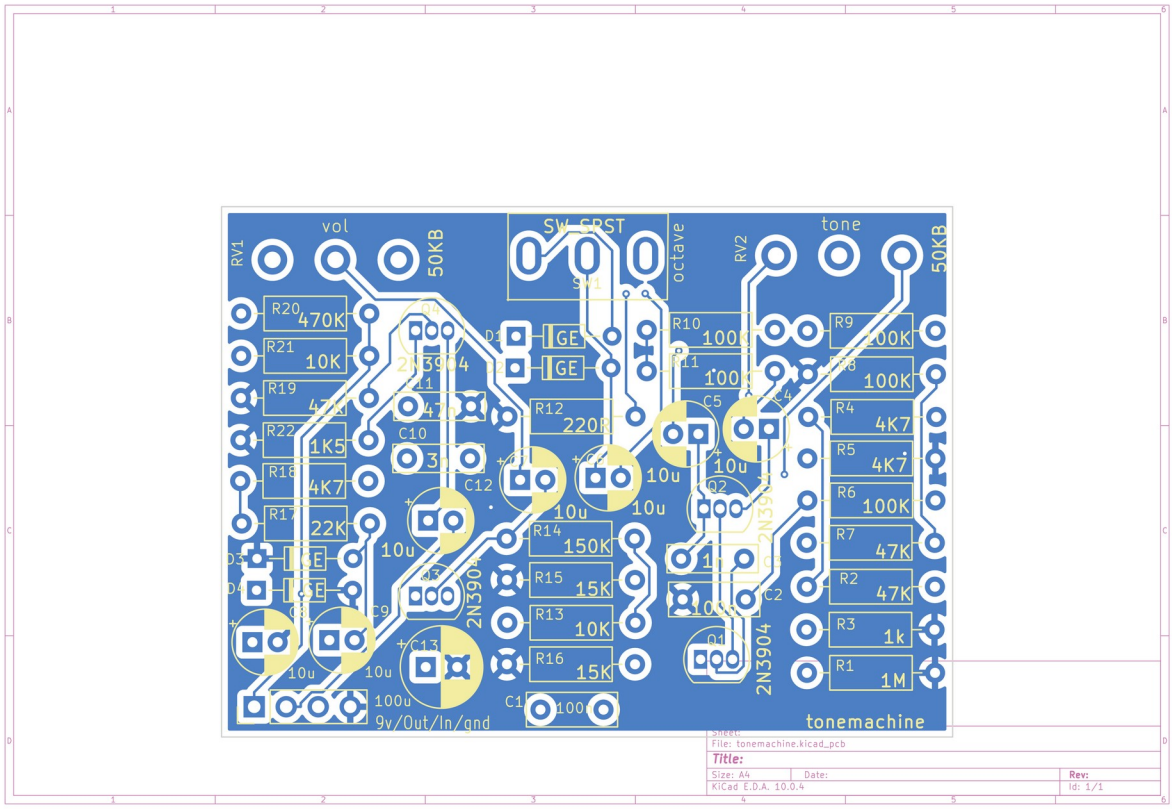
2. Schematic


Sheet: /
File: tonemachine.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1

3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

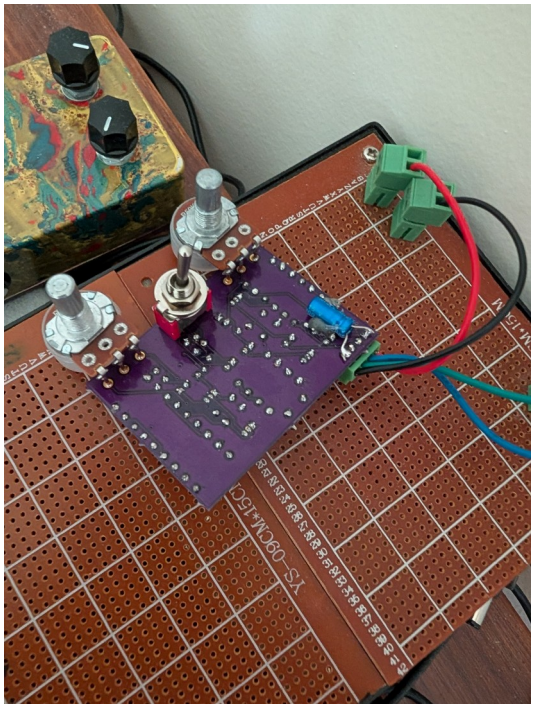
Reference	Value	Footprint	Qty	DNP
C1,C2	100n	Capacitor_THT:C_Rect_L7.0mm_W2.5mm_P5.00mm	2	
C3	1n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C4-C9,C12	10u	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm	7	
C10	3n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C11	47n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C13	100u	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm	1	
D1-D4	GE	Diode_THT:D_DO-34_SOD68_P7.62mm_Horizontal	4	
J1	Conn_01x04_Pin	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	1	
Q1-Q4	2N3904	Package_TO_SOT_THT:TO-92_Inline	4	
R1	1M	Resistor_THT:R_Axial_DIN02	1	

		07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal		
R2,R7,R19	47K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	3	
R3	1k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R4,R5,R18	4K7	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	3	
R6,R8-R11	100K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	5	
R12	220R	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R13,R21	10K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	2	
R14	150K	Resistor_THT: R_Axial_DIN02	1	

		07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal		
R15,R16	15K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	2	
R17	22K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R20	470K	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R22	1K5	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
RV1,RV2	50KB	pedals:Pot_Un derside	2	
SW1	spst	pedals:SWITC H-M1-SPDT	1	

Total line items: 23

Solder side.



Component side:

